D02 · Supply Planning & Scheduling

05 · Schedule adherence & OEE-driven scheduling

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1 WHAT IT IS

OEE-driven scheduling builds the plan against effective capacity — nominal capacity discounted by real availability, performance, and quality (OEE) and by the variability that makes effective throughput fall as utilization rises — and then measures schedule adherence, instead of scheduling against theoretical capacity the plant has never actually achieved.

2 WHY IT MATTERS

Kingman 1961 **(Kingman, 1961 — peer-reviewed)** proved that in any system with variability, waiting time — and therefore cycle time — rises sharply as utilization approaches 100%, diverging toward infinity at full load. Scheduling a plant to its nominal capacity assumes utilization can sit near 100% with no penalty; Kingman shows that is exactly where cycle time and queues explode. Layer on real OEE losses — downtime, speed loss, defects — and the gap between theoretical and achievable throughput is large and permanent. The result is chronic schedule misses: **a plant scheduled to 95% of nominal that actually runs at 65% OEE is committing to a number it cannot hit**, every cycle, and no one trusts a schedule the floor never makes.

3 HOW IT IS DONE

1. **Measure real OEE before scheduling to capacity.** Decompose availability, performance, and quality on the constraint resources. Nominal capacity is a nameplate; OEE is what the plant actually delivers, and the schedule must be built on the latter.
2. **Schedule to effective capacity, not nominal.** Discount nominal capacity by OEE and load the plan against that. A schedule built on theoretical capacity is a promise the floor was never able to keep, and its failure is arithmetic, not effort.
3. **Leave headroom for variability, especially at the bottleneck.** Kingman's law means cycle time explodes as the constraint approaches 100% utilization. Planning the bottleneck to run flat-out guarantees queues and tardiness; deliberate headroom is what keeps the schedule executable.
4. **Measure schedule adherence as a first-class KPI.** Track how much of the committed schedule was actually produced, in the planned sequence and quantity. Adherence is the truth test of the whole supply plan — a high-fidelity schedule no one follows is worse than a modest one they do.
5. **Feed adherence and OEE losses back into the next schedule.** If a resource consistently delivers 70% of plan, schedule it at 70% until the loss is fixed. Scheduling the aspiration instead of the reality just relocates the miss to next week.
6. **Attack the largest OEE loss to grow capacity.** The fastest capacity gain is usually closing the biggest OEE gap — a changeover, a chronic micro-stoppage, a quality loss — not adding equipment. OEE-driven scheduling and OEE improvement are the same program viewed from two ends.

> **WORKED EXAMPLE — ILLUSTRATIVE**
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> A plant scheduled its bottleneck line to 92% of nameplate — **baseline: actual OEE 64%, schedule adherence 71%, chronic tardiness and a floor that openly ignored the schedule**. It measured OEE, re-based the schedule on effective capacity with bottleneck headroom, and made adherence a tracked KPI. Results: **schedule adherence 71% -> 90%, on-time delivery up sharply, and a targeted attack on the largest OEE loss lifted effective capacity ~7%** — the schedule became credible because it was finally built on the capacity the plant actually had.

IN PRACTICE

Base the schedule on measured OEE

Use real, recent OEE by resource as the capacity input, refreshed as the plant changes. A schedule that loads to nameplate is structurally unachievable; one that loads to measured effective capacity is the foundation of adherence. The OEE measurement is the prerequisite, not a parallel improvement project.

Deliberate bottleneck headroom

Plan the constraint resource below full utilization on purpose, sized to the variability it faces. The headroom looks like waste on a utilization report and is in fact what protects cycle time and on-time delivery — running the bottleneck flat-out is the most common self-inflicted adherence failure.

Adherence on the S&OP scorecard

Put schedule adherence alongside forecast accuracy and plan adherence on the planning scorecard. It closes the loop from plan to floor and exposes whether the supply plan is connected to execution or quietly disconnected from it — the metric that keeps scheduling honest.

Target the largest OEE loss

Treat the biggest OEE loss as the capacity project it is. Closing a chronic changeover, micro-stoppage, or quality loss adds effective capacity faster and cheaper than capital, and directly raises the ceiling the schedule can commit to. OEE-driven scheduling makes the highest-value improvement target obvious.

WHAT YOU WOULD USE

SCHEDULING / OEE & MES

Blue Yonder / SAP (PP-DS, DM) / o9 — enterprise tier-1 — Finite scheduling that consumes effective-capacity and OEE inputs and ties adherence back to the plan. **(Blue Yonder, SAP, o9 — vendor)** The choice when scheduling, capacity, and the integrated plan must share one model and one set of capacity numbers.

MES / OEE platforms (Siemens, AVEVA, Rockwell) — Manufacturing-execution and OEE systems that measure real availability, performance, and quality and feed effective capacity to the scheduler. **(Siemens, AVEVA, Rockwell — vendor)** A strong fit where the priority is getting trustworthy OEE data into the scheduling input.

Dedicated OEE / shop-floor analytics specialists — Focused OEE-monitoring and shop-floor analytics tools that quantify and attack losses. **(specialized vendors — vendor)** The vertical option when the immediate need is OEE visibility and loss reduction more than integrated scheduling.

For operations that don't need a platform — A measured OEE figure per constraint resource, schedules loaded to effective capacity with bottleneck headroom, and adherence tracked in a spreadsheet. The decisive move is scheduling to the capacity the plant actually has — a discipline, not a purchase — with OEE measurement as the only real prerequisite.

4 SO WHAT

Operations that schedule to effective capacity with headroom produce a plan the floor can actually hit; operations that schedule to nameplate commit to a number Kingman's law and OEE losses make impossible, then call the inevitable miss a discipline problem. Chronic low schedule adherence is rarely a floor failure — it is a planning failure that scheduled the nameplate instead of the plant. The gap between nominal and effective capacity is not noise to push through; it is the schedule's honesty.

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